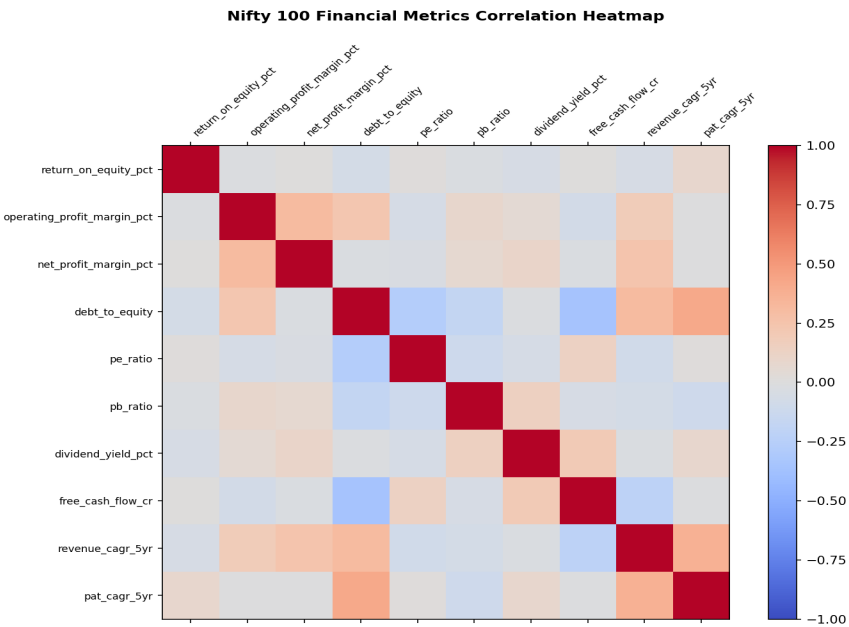


NIFTY 100 FINANCIAL INTELLIGENCE PLATFORM

COMPREHENSIVE ANALYST & SYSTEM USER GUIDE

Production Architecture, Quantitative Engines, Screener Presets & Dashboard Navigation

Document Version:	1.0 Production Ready
Target Universe:	92 Nifty 100 Index Constituents (10-13 Years History)
Computed KPIs:	50+ Fundamental, Growth, Cash Flow & Valuation Ratios
Primary Modules:	12 Interconnected Modules (ETL, Screener, Scoring, Reports, API, Dashboard)
Data Classification:	Confidential — Internal Use Only Data Analytics Division
Publication Date:	June 2026



1. Platform Overview & 7-Layer Architecture

The Nifty 100 Financial Intelligence Platform transforms raw, disparate annual filings and financial statements into high-conviction quantitative signals, institutional tearsheets, and interactive screener workflows. The platform operates on a robust 7-layer data platform architecture designed for determinism, auditability, and speed.

Layer	Components	Primary Technologies
L1: Ingestion	Excel loaders with header=1 detection, ticker validation	Python, openpyxl, pandas
L2: Normalisation	Fiscal year normaliser (YYYY-MM), deduplication engine	src/etl/normaliser.py
L3: Persistent Store	SQLite database with 10 tables, relational FKs, index trees	SQLite 3.x, data/nifty100.db
L4: Analytics Engine	50+ KPIs, CAGR engine, cashflow intelligence, capital patterns	NumPy, pandas, scipy.stats
L5: Intelligence Layer	0-100 Composite score, 11 peer percentiles, KMeans clusters	scikit-learn, rule engine
L6: Reporting Layer	92 tearsheets, 11 sector PDFs, portfolio master PDF	ReportLab, Matplotlib
L7: Interface & API	Interactive 8-screen dashboard, 16 REST endpoints	Streamlit, FastAPI, Uvicorn

Core Operating Principles:

- **Full Traceability:** Every calculated KPI is traceable to a specific mathematical formula and underlying statement row.
- **Graceful Edge-Case Handling:** Division by zero, negative shareholder equity, debt-free flags, and turnaround sign inversions are safely handled without data distortion.
- **Zero Data Leakage:** Annual metrics are strictly indexed by standardized fiscal year labels (YYYY-MM).

2. Data Warehouse & Schema Dictionary

The platform aggregates 7 core and 5 supplementary datasets spanning over a decade of audited corporate disclosures. All datasets are unified around the master NSE ticker primary key.

Dataset / Table	Type	Coverage	Key Columns & Purpose
companies	Core Master	92 constituents	Ticker id, legal name, logo URL, business profile, face value
profitandloss	Core Time-Series	FY 2010–2024	Sales, expenses, operating profit, OPM, net profit, EPS
balancesheet	Core Time-Series	FY 2010–2024	Equity capital, reserves, borrowings, total assets/liabilities
cashflow	Core Time-Series	FY 2010–2024	Operating activity (CFO), investing (CFI), financing (CFF)
sectors	Supplementary	92 companies	Broad sector (11 sectors), sub-sector, market cap category
market_cap	Supplementary	2019–2024	Market cap, enterprise value, P/E, P/B, EV/EBITDA, dividend yield
stock_prices	Supplementary	60 monthly rows	Monthly OHLCV price series (Jan 2020–Dec 2024)
peer_groups	Supplementary	11 peer clusters	Peer group name, company ID, benchmark constituent flag
financial_ratios	Computed Warehouse	1,155 records	50+ computed KPIs, 3Y/5Y/10Y CAGRs, composite quality score

3. Financial KPI & Ratio Calculation Reference

The ratio engine evaluates over 50 indicators across 4 key dimensions: Profitability, Solvency, Efficiency, and Cash Flow Quality.

KPI Category	Metric & Formula	Institutional Benchmark	Edge-Case Handling
Profitability	ROE: Net Profit / (Equity + Reserves) × 100	>15% Good, >20% Exceptional	Returns None if total equity <= 0
Profitability	OPM: Operating Profit / Sales × 100	>15% Stable, >25% Strong Moat	Cross-checked against reported OPM ±1%
Profitability	ROCE: EBIT / Capital Employed × 100	>15% Preferred	Sector-relative treatment for Banks/NBFCs
Solvency	D/E: Borrowings / Equity	<1.0 Safe, <0.5 Conservative	Zero debt yields exactly 0.0x
Solvency	ICR: (OP + Other Income) / Interest	>3.0x Safe, >5.0x Outstanding	Interest=0 flags 999.0 ('Debt Free')
Cash Flow	FCF: CFO + CFI	>0 Cash Generation	Negative allowed; 3-yr run flags risk
Cash Quality	CFO/PAT: Operating Cash / Net Profit	>1.0 High Quality, <0.5 Accrual Risk	Identifies aggressive revenue recognition
Efficiency	Asset Turnover: Sales / Total Assets	>1.0x Normal, >2.0x Asset Light	None if total assets = 0

4. Multi-Parameter Screener & 6 Preset Screens

The screener engine combines 18 configurable criteria to isolate investment candidates. Presets are loaded from `config/screener_config.yaml`.

Preset Name	Filter Logic	Primary Rank Metric	Target Universe
Quality Compounder	ROE > 15%, D/E < 1.0, FCF > 0, 5Y Rev CAGR > 10%	Composite Health Score	Elite franchise compounders (15–35 cos)
Value Pick	P/E < 20, P/B < 3.0, D/E < 2.0, Div Yield > 1%	FCF Yield	Undervalued cash generators (10–25 cos)
Growth Accelerator	5Y PAT CAGR > 20%, 5Y Rev CAGR > 15%, D/E < 2.0	5Y PAT CAGR	High velocity growth businesses (8–20 cos)
Dividend Champion	Div Yield > 2%, Payout < 80%, FCF > 0	Dividend Yield	Sustainable high-yield income (10–20 cos)
Debt-Free Blue Chip	D/E = 0, ROE > 12%, Revenue > ₹5,000 Cr	ROE %	Pristine zero-debt balance sheets (15–30 cos)
Turnaround Watch	3Y Rev CAGR > 10%, FCF positive, D/E improving	3Y Revenue CAGR	Operational turnaround candidates (5–15 cos)

Financial Sector Carve-Out Rule:

Per Risk Mitigation Rule R-04, companies classified under 'Financials' (Commercial Banks, NBFCs) are exempt from non-financial D/E screening ceilings. Their leverage is evaluated through sector-normalized return on assets and regulatory tier equity.

5. Financial Health Scoring Model (0–100)

The platform uses an institutional composite scoring framework that weights fundamental quality across 4 balanced dimensions.

Dimension	Weight	Sub-Metrics	Normalisation & Scoring Logic
Profitability	35%	ROE (15%), OPM (10%), NPM (10%)	Winsorised P10–P90, scaled 0–100 percentile rank
Cash Quality	30%	5Y FCF CAGR (15%), CFO/PAT (10%), FCF > 0 (5%)	Scaled cash generation; FCF > 0 awards 100 pts
Growth	20%	5Y Revenue CAGR (10%), 5Y PAT CAGR (10%)	Penalises negative base turnaround anomalies
Leverage	15%	D/E Score (10%), ICR Score (5%)	D/E: 0=100, 0.5=85, 1.0=70, 2.0=50. ICR: >10=100, >5=75

Composite Score Rating Tiers:

80 – 100	Excellent	Green (#2E7D32)	Market leading balance sheet, elite cash flows, low default risk
65 – 79	Good	Teal (#0288D1)	Solid fundamentals with sustainable growth and manageable leverage
50 – 64	Average	Yellow (#FBC02D)	Benchmark performance; mixed capital efficiency or moderate debt
35 – 49	Weak	Orange (#FB8C00)	Stressed cash generation or high leverage requiring active monitoring
0 – 34	Poor	Red (#D32F2F)	Severe operational deficits or structural financial impairment

6. Peer Comparison & Radar Analytics

The peer engine categorizes companies into 11 distinct industry peer clusters. It computes within-group percentile rankings for 20 metrics and generates individual radar chart profiles.

11 Dedicated Peer Groups:

1. **Private Banks:** HDFCBANK, ICICIBANK, AXISBANK, KOTAKBANK, INDUSINDBK
2. **Public Sector Banks:** SBIN, BANKBARODA, CANBK, PNB
3. **IT Services:** TCS, INFY, HCLTECH, TECHM, LTIM
4. **Pharmaceuticals:** SUNPHARMA, CIPLA, DRREDDY, DIVISLAB, TORNTPHARM
5. **Automobiles:** MARUTI, TATAMOTORS, M&M;, BAJAJ-AUTO, EICHERMOT, HEROMOTOCO, TVSMOTOR
6. **Life Insurance:** LIC, HDFCLIFE, SBILIFE, ICICIPRULI
7. **Oil & Gas:** RELIANCE, ONGC, BPCL, IOC, GAIL
8. **Power & Utilities:** NTPC, POWERGRID, TATAPOWER, ADANIPOWER, NHPC, JSWENERGY, ADANIGREEN
9. **Steel & Metals:** TATASTEEL, JSWSTEEL, JINDALSTEL, HINDALCO
10. **FMCG:** HINDUNILVR, ITC, BRITANNIA, DABUR, NESTLEIND, GODREJCP, TATACONSUM
11. **Consumer Finance:** BAJFINANCE, CHOLAFIN, SHRIRAMFIN

Radar Chart Interpretation:

Each radar chart visualizes a constituent's profile across 8 axes: ROE, OPM, NPM, 5Y Revenue CAGR, 5Y PAT CAGR, CFO/PAT Quality, FCF, and Composite Quality. The blue shaded polygon reflects company performance, while the orange dashed line shows the peer group average.

7. Cash Flow Intelligence & Capital Allocation Archetypes

Every company-year cash flow profile is classified into one of 8 capital allocation archetypes based on the mathematical sign of Operating Cash (CFO), Investing Cash (CFI), and Financing Cash (CFF).

Archetype Label	CFO	CFI	CFF	Strategic Interpretation & Health Implication
Reinvestor & Shareholder Returns	+	–	–	Ideal profile: Strong operating cash funding CapEx and paying dividends/debt
Growth via External Capital	+	–	+	Aggressive capacity expansion funded by both internal cash and external debt/equity
Divestment & Debt Repayment	+	+	–	Core operations positive; asset sales used to de-lever the balance sheet
Cash Hoarding	+	+	+	Accumulating liquid reserves across all channels; potential M&A; war-chest
Distress / External Funding	–	–	+	Red flag: Operations burning cash while CapEx and deficits are funded by debt raises
Rapid Cash Burn	–	–	–	Severe liquidity depletion: operating deficit, CapEx outlays, and net cash outflow
Asset Sale & Debt Funding	–	+	+	Emergency liquidity triage: selling fixed assets and raising loans to support operations
Restructuring & Contraction	–	+	–	Downsizing operations and liquidating investments while repaying senior lenders

8. Streamlit Dashboard Navigation Guide

The platform features an analyst dashboard accessible locally via streamlit run src/dashboard/app.py. The application is organized across 8 dedicated modules.

Screen / Page	Key Visualizations & Analytics	User Controls & Export Options
01 Home / Overview	Nifty 100 index summary KPIs, sector market cap donut, health band distributions	Sector filter chips, year selection dropdown
02 Company Profile	Company header card, 6 KPI tiles, 10Y Revenue/PAT bar charts, Balance Sheet breakdown, CF waterfall	Ticker search autocomplete, year range slider, PDF tearsheet download
03 Screener	Interactive table with 18 filterable metrics, live match counter, composite score ranking	Preset selector (6 screens), sidebar numeric sliders, CSV & Excel export buttons
04 Peer Comparison	Interactive Plotly radar chart, side-by-side comparison table, benchmark delta heatmap	Peer group dropdown, metric axis toggle, PNG chart save
05 Trend Analysis	10-year historical sparklines, YoY percentage change annotations, multi-metric overlay	Company search, multi-metric selector, CSV series export
06 Sector Analysis	Sector bubble chart (Sales vs ROE, bubble size = Mkt Cap), sector median KPI comparison	Sector selector, bubble axis parameters, PNG export
07 Capital Map	Treemap of 92 companies partitioned into the 8 capital allocation archetypes	Year slider, click-to-drill company list
08 Annual Reports	BSE India primary filing repository table with status flags and download links	Ticker search, filing year filter, direct BSE URL links

9. REST API Integration & Data Quality Rules

A FastAPI server exposes 16 production endpoints for programmatic integration. Start the API using `uvicorn src.api.main:app --port 8000`. Interactive OpenAPI documentation is available at <http://localhost:8000/docs>.

Endpoint Route	Method	Parameters	Description & Response Payload
/api/v1/health	GET	None	Health check returning SQLite row counts and uptime
/api/v1/companies	GET	sector, search	Constituent company list with base metadata
/api/v1/companies/{ticker}	GET	ticker	Comprehensive company profile with latest KPIs and pros/cons
/api/v1/companies/{ticker}/ratios	GET	ticker, year	Full time series of 50+ computed financial ratios
/api/v1/companies/{ticker}/tearsheet	GET	ticker	Streams pre-generated 2-page institutional tearsheet PDF
/api/v1/screener	GET	min_roe, max_de, sector	Returns ranked list of matching companies with composite score
/api/v1/sectors	GET	None	Sector medians for ROE, P/E, D/E and total market cap
/api/v1/portfolio/stats	GET	None	P10 to P90 portfolio distribution benchmarks across all KPIs

Data Quality (DQ) Enforcement:

16 programmatic validation rules ensure data warehouse integrity on every ETL run. Zero critical violations are permitted for production sign-off. All anomalies are permanently logged to `output/validation_failures.csv`.